

ZSWIM8-6 ■■■■■ · Grant Writing Master Template

"Energy-stress-driven metabolic memory via ZSWIM8-dependent miRNA decay"

■■■■■■■■ → ■■■■■■■■ (K99/R00 · F32 · R01 · ■■■■ · RGC ECS · NRF)

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schematic(■■■/■■), ■■■■■■■■■■ (a) ■■■■■■■■■■ PSSM ■■■■ (b) ■■■■■■ (c)

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1.1 Specific Aims (■■■■■■■■■■)

[Opening hook] MicroRNAs, once regarded as constitutively stable, are actively destroyed on a physiological timescale by target-directed miRNA decay (TDMD), in which the ZSWIM8 ubiquitin ligase recognizes a highly complementary target engaged by an AGO2–miRNA complex and marks AGO2 for degradation.

[Knowledge gap] Although the ZSWIM8 TDMD machinery is now structurally defined, whether and how it is tuned by the cell's metabolic state is unknown. In particular, whether the energy sensor AMPK—activated within minutes of nutrient stress—directly controls the rate of miRNA turnover has never been tested. This gap is pressing because type 2 diabetes (T2D) exhibits "metabolic memory": tissues retain a disordered metabolic program long after glycemic control is restored, yet no molecular mechanism explains this persistence.

[Long-term goal & central hypothesis] Our long-term goal is to define the metabolic control of miRNA stability and exploit it therapeutically. **Our central hypothesis is that energy stress, through AMPK, phosphorylates ZSWIM8 at S608, accelerating TDMD of specific metabolic miRNAs and thereby encoding a durable, reversible metabolic memory.** This hypothesis is grounded in our transparent AMPK substrate-motif scan, which places ZSWIM8-S608 in a textbook AMPK recognition context (...KRR-S608...) within a disordered, kinase-accessible region.

[Aim 1 — molecular event] *Test whether AMPK directly phosphorylates ZSWIM8 at S608. In vitro kinase assays with recombinant AMPK and ZSWIM8 fragments, phospho-specific detection, and S608A/S608D mutants will establish direct phosphorylation and its dependence on the predicted motif. (Predicted outcome: S608 phosphorylation is AMP-dependent and abolished by S608A.)*

[Aim 2 — functional readout] Determine how S608 phosphorylation sets the rate of TDMD. Using time-resolved small-RNA sequencing after transcriptional shut-off in ZSWIM8-reconstituted cells (WT / S608A / S608D / KO), we will fit decay-rate constants for candidate miRNAs. (Predicted ordering: S608D > WT > S608A > KO.)

[Aim 3 — disease / memory] *Test whether AMPK-driven ZSWIM8 activity encodes metabolic memory.* In a T2D pig model and in fibrosis organoids subjected to a defined nutrient-stress pulse, we will ask whether a transient stress produces a durable, S608-dependent shift in metabolic miRNA levels and phenotype. (*Predicted: transient stress → lasting, S608-dependent miRNA suppression.*)

[Payoff] Successful completion will establish the first metabolic input to the TDMD machinery, define a phospho-switch (ZSWIM8-S608) as a druggable node, and open metabolic control of miRNA stability as a new axis in diabetes and its complications.

1.2 Significance (R01/K99 ■■■;fellowship ■■)

- **Clinical burden.** Metabolic memory drives the persistence of diabetic complications despite glycemic control; there is no therapy that erases this memory. A metabolic lever on miRNA decay would be a mechanistically novel entry point.
- **Mechanistic advance.** Establishing AMPK as an upstream regulator of TDMD would convert miRNA turnover from a housekeeping process into a signal-responsive, druggable pathway.
- **Conceptual reach.** Because ZSWIM8 governs decay of many miRNAs, a metabolic input would have consequences across the miRNAome, not a single miRNA.
- **Unique resources.** Direct access to a T2D pig model, intestinal/cardiac organoids, a lab-native small-RNA sequencing pipeline, and base/prime editing positions us to test this axis in human-relevant, large-animal contexts other labs cannot easily replicate.

1.3 Innovation (R01/K99 ■;fellowship ■■■ Significance)

- **Conceptual:** a paradigm shift—miRNA decay is not constitutive but is set by metabolic state through a defined kinase→ligase axis.
- **New target:** ZSWIM8-S608 as, to our knowledge, the first phosphosite proposed to couple energy sensing to TDMD.

① ■■■■■■ — ■■■■■■: - NIH/RGC/NRF(■■■■■): ■ 1.1 ■ TDMD ■■■■(■■■■)■ -
■■■■(■■■■): ■■■■ 2 ■■■■■■, ■■■■■■ ■■■■■■ ■■■■■■

② ■■■■■■ — ■■■■■■:■■■(■■■):We hypothesize that energy stress, through AMPK, shifts the rate of ZSWIM8-dependent miRNA decay...(■ shift/hypothesize,■■) - ■■■(■■):Our data demonstrate...; we therefore hypothesize that AMPK phosphorylates ZSWIM8 at S608...(■ demonstrate,■■)

③ ■■■■■ — ■■■■: - K99/F32:■■■"■■■■■■■■■■ X"→■■■■■■■■■■ - ■■/NRF:■■■"■■■■■■ X + ■■■■ Y"→■■■■■■■■■■

2.3 ■■■■■(■■■■■)

1. **Specific Aims** (1.1) — , ,
2. **Career/Training** (K99/ /NRF ; R01)
3. **Approach** : predicted; predicted
4. **NOFO**: SABV/Rigor

[illegible]